Assignment 3 — MVP Implementation & Demo

Minh Quan Dao (qd66) · Interactive Diffusion-Based 3D Generation Studio · Pillars: **Procedural Generation + Rendering**

Source: https://github.com/quandao10/diffusion-3d-studio · Web build: https://quandao10.github.io/diffusion-3d-studio/ (static) · Live studio: public GPU link (while the server runs) · Video: https://www.youtube.com/watch?v=UGFzpFrmWcE

What runs. A text prompt is turned into an explorable 3D object by optimising **3D Gaussians** with gradients from a **frozen diffusion model** (official DreamGaussian code, commit `69a3c70`, on an NVIDIA A100). The result is rendered in real time by my WebGL2 splatting viewer inside a public web studio: anyone can submit a prompt, watch the optimisation live, and orbit the result.

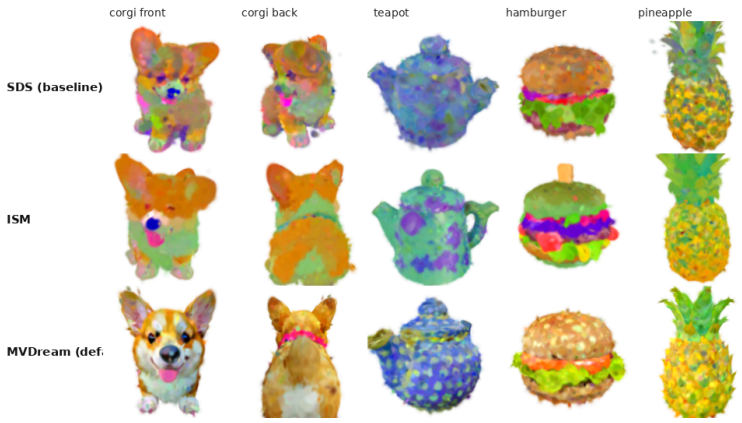
1. Plan vs. Reality

A2 MVP item	Status	What was built / evidence
Text prompt → frozen diffusion prior → score distillation (SDS)	**done**	SD 2.1-base, CFG 100, annealed t; gradients reach all Gaussian parameters at every step (verified, `metrics.jsonl`)
NeRF + differentiable volume rendering	**changed**	explicit 3D Gaussians + differentiable CUDA rasterizer: ≈47 s per 500-step run instead of hours, so I could iterate
Interactive novel-view viewer; progress view; seed/guidance/steps controls	**done**	own WebGL2 viewer, live loss/gradient/#Gaussian charts, snapshot scrubber, method picker
COULD: 3DGS, mesh export, variations	**done**	3DGS is the core; textured OBJ via upstream export (CLI); public community gallery
COULD: VSD comparison	**changed**	VSD (LoRA training, hours) replaced by fast alternatives CSD, NFSD, ISM + MVDream, measured (Sec. 2)
Stage-2 mesh refinement; production UI	**cut**	not run; the studio is a working prototype, not a polished product

2. Improving DreamGaussian: faster / higher-quality guidance

I added CSD, NFSD and Interval Score Matching (LucidDreamer) as alternative distillation losses on the same frozen SD 2.1, with view-dependent prompts (`scripts/guidance_plus.py`), and enabled upstream's MVDream multi-view guidance. All five were compared on 5 prompts, seed 0, 500 steps, one free A100 per run. CLIP = ViT-L/14 prompt–view similarity over 8 views (`evidence/guidance_comparison*.csv`).

guidance	CLIP	optimise	total	peak VRAM	visual result
SDS (upstream)	26.38	47 s	57 s	4.4 GB	blotchy, over-saturated; corgi gets a 2nd face (Janus)
CSD	24.65	51 s	60 s	4.4 GB	noisy, speckled
NFSD	25.45	53 s	63 s	4.4 GB	natural colours, blurry
ISM (LucidDreamer)	25.95	107 s	115 s	4.4 GB	Janus fixed, colours off
MVDream (default)	25.94	91 s	127 s	10.7 GB	**coherent, no Janus, sharpest**



Same seed and steps. SDS gives the corgi a second face; MVDream is coherent and sharp.

MVDream step sweep (CLIP / total time):

300 steps	26.4	86 s	clean, softer
500 steps	25.9	127 s	**sharpest**
800 steps	25.9	193 s	over-saturated
1200 steps	23.2	283 s	blown out

CLIP barely separates the methods (it rewards saturated colours), so I judged visually. **MVDream is the default** (~2 min per object). 500 steps is the sharpest and longer runs over-saturate, so 500 is also the public cap.

3. Stress test (measured, CUDA rasterizer, dedicated A100)

Forward render of a 311k-Gaussian result along a fixed 360° orbit (120 frames, warm-up excluded, synchronised timing, `evidence/performance.csv`). **FPS vs resolution:** 256^2: 315 · 512^2: 368 · 1024^2: 175 · 2048^2: 58 · 4096^2: 17. **FPS vs Gaussian count at 512^2** (k jittered clones): 0.3M: 354.5 · 1.2M: 144.6 · 5.0M: 44.8 · 19.9M: 11.7 · 39.9M: 5.5; peak VRAM 23.9 GB at 40M; **CUDA out-of-memory at 80M** (failure point). The A2 viewer target (≥30 FPS) holds up to 2048^2 and ≈5M Gaussians. The browser viewer is also pushed ×1–×16 live in the video (it rendered on a GPU shared with another user's job, so those numbers are indicative only).

4. Public studio (runtime system)

- **Scheduling:** each job runs on a GPU that has *no process of anyone* on it, so the number of parallel jobs follows the machine live (4 free A100s → up to 4 jobs). Otherwise jobs queue FIFO.
- **Visitors:** see free GPUs and an **ETA before submitting** (measured s/step × steps + overhead, simulated over the queue). Each session can have one job in progress, which it can **stop** to release the GPU. Finished models join a shared gallery.
- **Safety:** server bound to localhost behind an outbound Cloudflare tunnel; prompt filter, ≤500 steps, no visitor access to the stress test; server paths hidden from visitors' logs; scheduler and limits covered by in-process tests.

5. AI Utility

- **Runtime AI:** frozen SD 2.1-base and MVDream supply every gradient; nothing is trained. The original `stabilityai` SD repo now returns HTTP 401, so I used a mirror of the same weights (documented).
- **Coding agent (Claude Code), as planned in my tech spec:** very useful. From the logs, the environment build started 11:44 and the first successful 500-step run finished 12:04. It diagnosed: a missing `<cfloat>` in simple-knn; CUDA headers stored only under `targets/` on this server; a 37 GB peak-VRAM reading caused by `cudnn.benchmark` (fixed to 4.4 GB, aborted log kept); and headless-Chrome WebGL context loss. It also wrote most of the wrapper, guidance losses, viewer, studio, scheduler and benchmarks.
- **Cost:** every claim still had to be checked against renders and logs. CLIP alone would have picked the wrong method, and a blank viewer turned out to be a lost GPU context, not a shader bug. Time saved was not measured against a baseline. The video narration is synthetic (Kokoro TTS) and matches the on-screen captions.

6. Roadmap to Final

- Quality: combine MVDream with ISM-style deterministic targets, tune timestep/CFG schedules, add stage-2 mesh refinement and a mesh/Gaussian toggle.
- Studio: stable hosting (named tunnel or university reverse proxy), seed/method side-by-side compare, moderation for the shared gallery.
- Viewer: GPU sorting and level-of-detail for multi-million-Gaussian scenes; measure it on a dedicated GPU.

Build/run: Linux + NVIDIA GPU: `bash scripts/setup_env.sh`, then `python server.py` or `bash scripts/serve_public.sh` (README, `evidence/`). Demo video: 177.3 s screen capture of the running studio; only waits are fast-forwarded and labelled.